2022“CCF-蚂蚁绿色计算专项科研基金”

项 目 名 称

基于低耗能网络结构搜索优化技术的绿色搜推营模型研究

申报课题：13. 碳效率约束下的复杂 AI 模型的自动网络结构搜索优化

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1. 项目介绍

1.1. 题目

基于低耗能网络结构搜索优化技术的绿色搜推营模型研究

1.2. 背景及研究意义

搜推营业务作为多数互联网产品的主要流量入口和收入来源，有着大量的现实需求。随着深度学习技术在各大公司的普及和落地，以精度为目标的大模型在搜推营业务中得到了广泛的应用。然而，大模型同时也带来了训练时间长、推理能耗大等“绿色”负担。这些负担在企业层面表现为高额的能源支出，在社会层面表现为严重的环境污染，因此如何构建绿色高效的搜推营模型是企业和社会都极为关注的问题。为了构建“绿色”的搜推营模型、提升企业的绿色效能，本研究拟结合自动网络架构搜索(Neural Architecture Search，简称NAS)技术，提出通用且高效的模型剪枝与搜索优化方法，一方面大幅减小模型训练时长，另一方面降低模型的线上推理能耗。

当前，已有少量研究关注到搜推营模型的“绿色”问题。比如，Adnan等人[1]的研究中发现在淘宝电商数据集上训练一个模型，需要平均能耗56.39W的4块GPU同时训练621分钟。来自Facebook的研究者们[2]则发现产品级深度推荐模型的参数量在近些年以百亿级别增长，为了驱动这些模型的训练和推理甚至需要上百个GPU同时工作。由此可见，减小搜推营模型参数量是实现绿色计算目标的关键。

NAS技术作为搜索最优网络结构的一种方法，近年来在搜推营模型上得到了广泛研究[3]。通常，NAS利用验证集损失对模型结构进行调整，其目标函数可以形式化为一个双层优化的问题：

$$\min_{\mathcal{A}} \mathcal{L}_{valid}\left(\mathcal{W}^*(\mathcal{A}), \mathcal{A}\right)$$

$$s.t. \mathcal{W}^*(\mathcal{A}) = arg \min_{\mathcal{W}} \mathcal{L}_{train}\left(\mathcal{W}, \mathcal{A}\right)$$

其中，$\mathcal{L}_{train}$和$\mathcal{L}_{valid}$分别为训练集和验证集上损失，$\mathcal{A}$和$\mathcal{W}$分别为搜推营模型的参数和结构权重。当前，有两大类基于 NAS 的研究可以缓解搜推营模型参数量大的问题：①Embedding 维度搜索方法和②自动特征选择方法。Embedding 维度搜索方法寻求为每一个特征域搜索最优且最小的 embedding 维度。例如，ATML[4]和 PEP[5]提出了一种基于梯度的剪枝策略，自动化地为每一个特征域搜

索最优维度。但是， 这两种方法面临着搜索空间大的问题。为了解决该问题，Zhao 等人提出通过将 embedding 切分为多个片段， 从而减小搜索的方法—AutoEmb[6]和 ESAPN[7]。其中，AutoEmb 为每个 embedding 片段分配一个可训练的权重，通过加权的方式实现不同片段的组合。而 ESAPN 提出了一个基于特征频率的硬选择策略。自动特征选择方法则通过搜索最优的特征组合，减少冗余的特征输入 ， 从而达到压缩模型的目的。这类方法的代表有 MARLFS[8]和 AutoField[9]。其中，MARLFS 将特征选择看作为一种多智能体问题，提出了一种基于强化学习的方法。AutoField 则设计了一个控制网络来计算某个特征保留和选择的概率。

然而，现有NAS技术和绿色计算方法存在如下的几个问题与难点：

（1）搜索过程“不绿色”。NAS算法搜索过程中最重要的一个环节是对网络结构进行评估，如上述双层优化问题所示。现有的模型评估方法基本遵循训练集上训练-验证集上评估的两阶段范式，该过程中模型训练对资源的消耗巨大。此外，NAS需要同时对众多不同的网络结构进行评估，进一步加剧了搜索过程不“绿色”的问题。并且，搜推营业务本身具有场景任务复杂、时效性要求高等特点，即要求模型更新迭代频次高。然而，频繁地调用NAS算法会进一步放大其模型评估消耗大的劣势。因此，在搜推营业务中如何在搜索过程中减小每次模型评估的资源消耗，是真正实现NAS算法落地的关键所在。

（2）搜索结果不“绿色”。现有的NAS方法大多数依旧以提升模型的性能为目标，无法保证搜索得到的模型参数量更小、运行更高效。以上述的双层优化式为例，其优化目标为最小化验证集上的损失。然而，简单地加入模型复杂度作为优化目标，一方面会使得优化式复杂、解空间增大，另一方面会导致模型的性能下降。因此，如何应用NAS算法得到“绿色”模型的同时，又能保证模型的性能不显著下降，是将NAS应用到绿色计算的一个难点。

针对以上NAS技术和绿色计算存在的问题与难点，本研究将从NAS算法设计和绿色模型搜索两个方面攻克现实业务中绿色模型难搜索、难部署、难应用的问题，拟针对NAS算法设计低成本的模型评估策略、通过在NAS优化过程中加入合适的资源限制条件来获取绿色模型、构建Single-Shot或Zero-Shot的模型剪枝方法。

1.3. 研究目标

（1）结合搜推营模型结构的特性，设计：①高性能、低能耗的NAS算法，减少搜索模型结构时产生的碳排放；②搜索出绿色搜推营模型结构，能耗下降50%，模型效果下跌1%内；

（2）输出CCF-A类1-2篇学术论文；

（3）1篇核心专利，在蚂蚁3+业务场景落地上线；

1.4. 研究方法

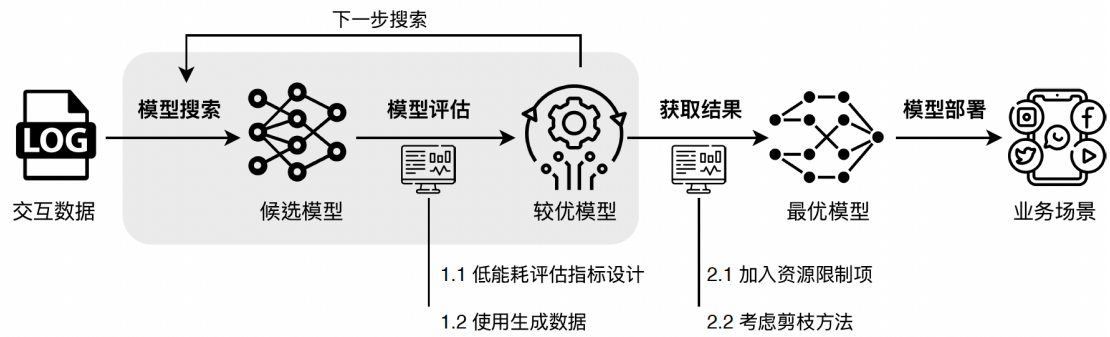

图 1 技术路线概览

搜广营业务场景的NAS算法运行流程可以表示为图 1。NAS算法（灰框部分）首先基于收集到用户的交互数据进行搜索，得到若干个候选模型；然后对这些候选模型的效果进行评估，根据评估结果得到较优的模型，从而继续进行下一步搜索，使搜索阶段的结果可以向模型表现提升的方向发展；达到NAS算法的停止条件后，我们获得搜索结果（最优模型）并部署到业务场景中。可以发现，在一次NAS算法运行过程中，模型评估过程反复进行，当候选模型数量多、需要运行NAS算法的频率高时，相对搜索过程，评估过程将消耗大量碳排放；同时，输出的最优模型只考虑效果的最优性，忽略了可能随之而来的高能耗。针对这两个方向（搜索过程“绿色”、搜索结果“绿色”），本项目拟采用以下技术路线：

1. 针对NAS算法设计低成本的模型评估策略

现存的NAS算法普遍采用参数共享（Parameter Sharing）策略[1]进行搜索结果的表现评估， 相比最初的方法已经大大减少了碳排放。具体而言， 以计算机视觉（CV）领域的NAS算法为例，采用参数共享策略将原来需要3000 GPU Hours[12]才能完成的NAS任务缩短至10 GPU Hours左右[11]。这启发我们高效的模型评估

策略设计可以帮助NAS算法保持甚至改进性能的同时大大减小能耗。

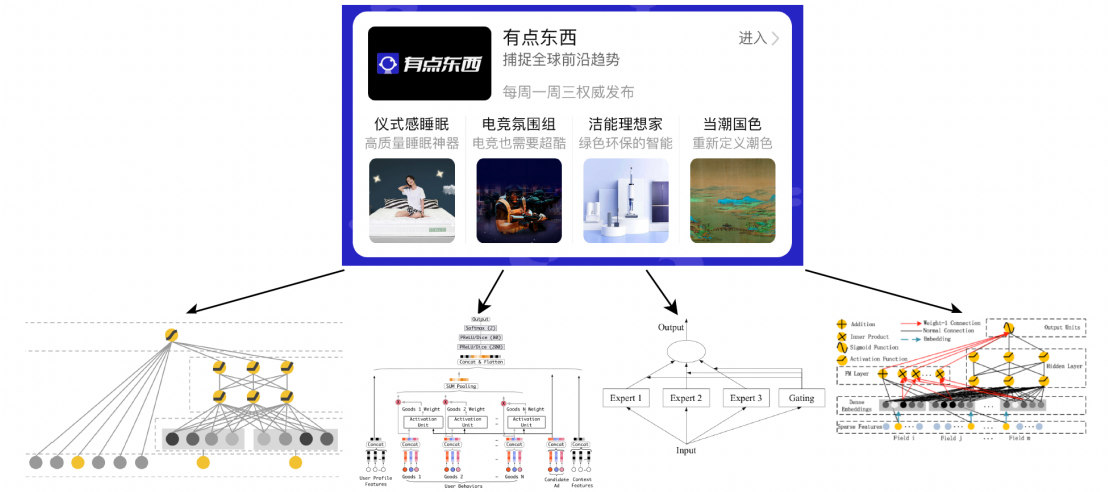

图 2 “有点东西推荐场景”

然而，由于搜广推模型需要适应不同的任务、场景，小时级别的模型搜索时间在重复完成搜索任务时仍然可能带来大量的碳排放。以“有点东西”的推荐场景为例，该频道每周一周三发布新主题，向用户推荐相关商品，由于不同主题数据分布不同、可以使用的用户、物品特征字段不同（如，“洁能理想家”产品中的“是否需要连接电源使用”属性不适用于其他主题），我们需要对应地使用不同结构的神经网络结构，则每周需要进行两次NAS搜索来获得不同的模型结构，小时级别的计算消耗在这样的频率下仍然会消耗大量碳排放。

1.1 不需要训练模型的评估策略

现存评估策略消耗的计算资源主要用于训练搜索得到的模型，获得其相对准确的预期效果（准确率等），从而使NAS算法最终得到高性能的模型。我们可以结合搜广营模型、场景的特点，设计不需要训练就可以得到的，且和模型实际效果正相关的指标（模型得分），达到高效评估模型的目的。举例来说，可以认为随机初始化的网络对数据的分辨能力能够部分地反映该网络训练完成后的预测效果。我们可以使用一种基于线性激活层（ReLU，Rectified Linear Unit）输出状态判断网络分辨能力的方法，将搜索网络所需的时间从小时级别降低到秒级且保持搜索结果的性能。

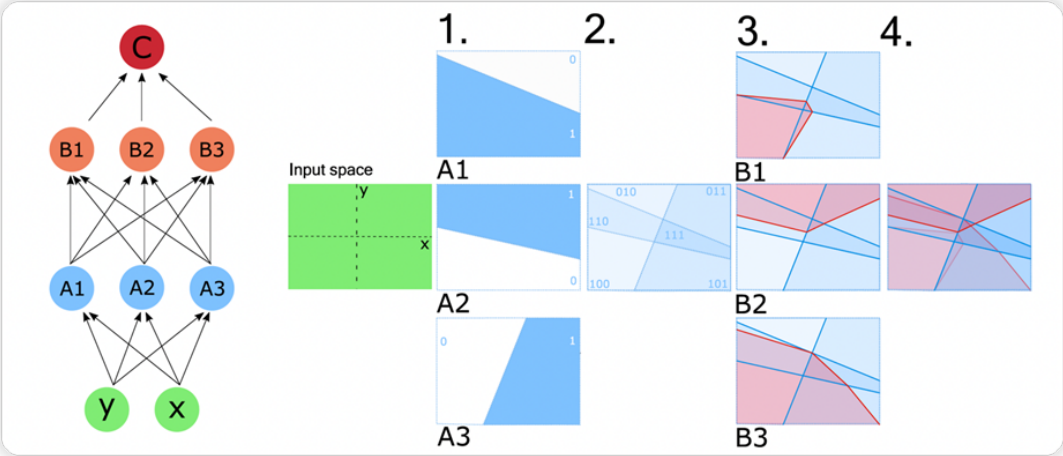

图 3 根据 ReLU 输出进行编码示意图

具体来说，如图 3所示，每一个线性激活层ReLU将整个空间分为两个部分，可以使用0、1对输出空间分别进行编码。如果同一层网络有多个ReLU激活函数，则该层的输出将被分为更多区域，然后将所有ReLU的输出分别编码后连接，作为该层最终的编码。例如图中，A层有3个ReLU，根据每个ReLU对空间的分解分别编码后相连，最终得到一串形如“010”的编码。经过B层后，再将该层得到的编码拼接于前面的编码后。例如，图中B层也有3个ReLu，因此示意网络的输出可以被编码为一串长度为6的二元向量（取值0或1）。

假设网络中ReLU数量为N_c，随机输入一批训练数据（规模为N），根据上述编码方式将所有训练数据对应的输出编码为长度为N_c的二元向量$c_i, i \leq N$，并用汉明距离$d_H(c_i, c_j) = \sum|c_i - c_j|$ 来衡量输出之间的相似度：

$$K_H = \begin{pmatrix} N_c - d_H(c_1, c_1) & \cdots & N_c - d_H(c_1, c_N) \\ \vdots & \ddots & \vdots \\ N_c - d_H(c_N, c_1) & \cdots & N_c - d_H(c_N, c_N) \end{pmatrix}$$

最后使用的得分函数为：$S = log|K_H|$,即矩阵K_H的对数行列式，越大的S反映了网络更强的分辨能力，因为输出编码之间的相似度小。利用该方法来评估NAS算法每次搜索得到的模型，可以避免评估过程中对候选模型的训练过程，大大减少需要的计算时间（小时级到秒级）。在推荐模型中，ReLU或类似的结构（PReLu、LeakyReLU）被广泛使用，因此该方法具有较好的使用前景。

1.2 使用生成数据对搜索阶段产生的候选模型进行评估

在推荐数据集中总会存在一些用户由于好奇、失误而造成的点击记录（噪声），无法反应用户的真实兴趣，且这些噪声很难被分辨出来[13]。同时，收集的交互

数据中无法避免地会存在特点相似的数据，遍历所有的数据进行模型训练或评估往往不是必要的[14]。而NAS搜索阶段的模型评估需要遍历一次验证集，当数据量较大、网络较复杂时，反复进行模型评估消耗的碳排量大大增长。我们可以设计生成模型来产生小体量但能反应数据整体分布的生成数据来进行模型效果的评估，从而显著提升碳效率。具体来说，将使用的生成模型表示为G，训练集表示为D_{train}，生成数据表示为D_G，$D_G = G(D_{train}; S)$，S表示指定的数据集规模。为了得到G，我们可以将训练目标构建为：

$$\min_{\mathrm{D_G}} \mathcal{L}_{val}(w^*|D_{train}) - \mathcal{L}_{val}(w^*|D_G)$$

其中$\mathcal{L}_{val}$表示推广营模型在验证集上的损失。通过这个优化目标，我们希望根据少量生成数据训练出的推荐模型（$w^*|D_G$）效果可以接近使用真实数据训练出的推荐模型（$w^*|D_{train}$）。得到生成模型G后，我们再通过控制数据集规模S的大小，生成模型评估阶段使用的小量数据集。举例来说，常用的推荐数据集Criteo共有4500万条交互记录，以8:1:1的比例切分数据集后，验证集中有450万条交互记录。如果需要得到精确的模型评估结果，我们需要遍历整个验证集。使用10万规模乃至1万规模的生成数据进行评估，可以节约90%以上的评估成本。

<u>讨论</u> 除了本方案提出的设计低成本模型评估策略之外，优化并控制搜索空间规模、限制候选模型数量是完成NAS搜索过程绿色化另外一个可能方向。但该方案存在以下缺点：1. 较难平衡NAS算法的效率和性能。搜广营模型中可供选择的计算单元（FM层、线性层、注意力层）较少，如果在此基础上为了提高碳效率放弃使用或限制使用较多的候选计算单元，虽然可以大大减少搜索中产生的候选模型数量，但过多的限制将导致输出模型的网络结构单一，能够取得的预测效果显著下降。如果为了保持输出模型预测效果而减少对搜索空间的限制，采用这种方法又很难达到与本方案所提出的从评估策略角度优化的方法相当的碳效率；2.在搜广营场景中的通用性差。搜广营场景中，除了上层模型计算单元的结构搜索之外，嵌入特征矩阵、特征交互的搜索也是重要的研究方向[4-7]。对于这两个问题，搜索空间往往是固定的。具体而言，对于嵌入特征矩阵的搜索，搜索空间是$\boldsymbol{E} \in R^{M\times d}$，即整个初始的特征矩阵，$M$由数据集决定，而$d$是需要搜索（针对嵌入矩阵维度的搜索）或者人为指定的参数（针对M的搜索）；对于特征交互的搜索，无论是对低阶交互（2阶）还是高阶交互（3阶及以上），搜索空间则是全

部可能的特征交互选择。因此，减小搜索空间的方法无法应用在这些问题上。本方案提出的通过设计低成本评估策略达成搜索绿色化的方案，由于评估候选模型是NAS搜索中不可以或缺的环节，因此通过修改原始NAS算法中的对应评估方法，本方案可以与任何NAS算法、搜广营模型中的NAS任务结合，有较强通用性。

2. 获取绿色的模型搜索结果

2.1 在NAS的搜索过程中加入资源限制项

大部分NAS算法的目的是为了得到高性能的网络结构，而没有考虑搜索结果的复杂度。以DARTS为例[13]，该算法的目标为：

$$\min_{\theta} \mathcal{L}_{val}(w^*(\theta), \theta)$$

$$\text{s.t. } w^*(\theta) = \underset{w}{\operatorname{argmin}} \mathcal{L}_{\text{train}}(w, \theta)$$

其中，θ表示各个候选模块的权重，w表示候选模块的参数。可以发现DARTS的目标是在所有模块充分训练后（w^*）选择最优的模块结构（θ），没有考虑最后得到的网络的复杂度。一种直观的方法为通过增加限制条件来控制搜索结果需要消耗的计算资源：

$$min_{\theta} \mathcal{L}_{val}(w^*(\theta), \theta)$$

$$\text{s.t. } w^*(\theta) = \underset{w}{argmin} \mathcal{L}_{\text{train}}(w, \theta),$$

$$\boldsymbol{C_L} \leq \boldsymbol{\Phi}(\theta) \leq \boldsymbol{C_H}$$

$\boldsymbol{\Phi}(\theta)$是一个多元函数，将参数$\theta$映射到具体的资源限制条件上。举例来说，常考虑的资源限制条件有两个：搜索得到的网络参数量、网络需要的浮点数运算次数。增加的限制条件$\boldsymbol{C_L} \leq \boldsymbol{\Phi}(\theta) \leq \boldsymbol{C_H}$，既可以控制使用资源的上限，又可以避免搜索结果使用的资源太少，导致模型效果显著下降。

考虑到$\boldsymbol{\Phi}(\theta)$可能是非凸的，我们可以将NAS任务分为两个阶段，第一个阶段中完成没有资源限制的搜索，在第二个阶段中，再根据资源限制条件得到一个接近最优结果的模型。这样的一步逼近需要消耗的计算资源显著小于无限制的搜索，因此整个过程需要消耗的资源接近DARTS，达到在不增加资源消耗的前提下完成绿色模型搜索的目。

在搜广营模型中，如果只从参数的角度考虑，嵌入表征矩阵往往占用了绝大部分的参数，我们在模型搜索的过程中对嵌入表征矩阵的稀疏性进行限制（如不

同特征字段使用的嵌入表征维度）就可以在满足资源消耗条件下找到最优的嵌入矩阵设计，包括特征选择（嵌入矩阵的行）、嵌入表征维度（嵌入矩阵的列）。从FLOPS角度考虑，则可以控制后续的模型结构（DNN，DIN，Transformer等）满足绿色要求。

2.2 Single-Shot或Zero-Shot的模型剪枝方法

NAS算法完成的是从无到有构建网络结构的过程，而模型剪枝的方法则可以完成现有模型的轻量化 ， 同样可以达到搜索绿色新模型的目的。使用Single-shot或zero-shot的剪枝方法，既满足搜索算法低能耗的要求，得到的轻量化模型参数、计算量也可以通过参数控制。举例来说，我们将剪枝问题构建如下：

$$min_{c,w} L(\boldsymbol{c} \odot \boldsymbol{w}; \mathcal{D}) = \min_{c,w} \frac{1}{n} \sum_{i=1}^{n} \mathrm{l}\big(\boldsymbol{c} \odot \boldsymbol{w}; (x_i, y_i)\big)$$

$$\text{s.t.}\ \boldsymbol{w} \in R^m, \boldsymbol{c} \in \{0,1\}^m, |\boldsymbol{c}|_0 \leq \kappa$$

其中，$\mathcal{D}$是训练数据集， $\boldsymbol{w}$ 是规模为m的待剪枝参数，$\boldsymbol{c}$为对应的mask，取值为1或0，分别表示参数被保留或舍弃。$\odot$表示Hardamard积，$\boldsymbol{c} \odot \boldsymbol{w}$ 代表剪枝结果。κ用于控制剪枝后保留的最大参数量。然后，通过对比某个参数缺失时对模型效果的影响来衡量该参数的贡献：

$$\Delta L_j(\boldsymbol{w}; \mathcal{D}) = L(\boldsymbol{1} \odot \boldsymbol{w}; \mathcal{D}) - L\left(\left(\boldsymbol{1} - \boldsymbol{e_j}\right) \odot w; \mathcal{D}\right)$$

$\boldsymbol{e_j}$ 是只有第j个元素为1，其余均为0的单位向量。$\Delta L_j(\boldsymbol{w}; \mathcal{D})$表示了第$j$个参数被舍弃对loss造成的影响。计算所有的$\Delta L_j(\boldsymbol{w}; \mathcal{D})$会消耗大量的计算资源，因此，我们可以进一步对$\Delta L_j(\boldsymbol{w}; \mathcal{D})$进行近似，并将$\boldsymbol{c}$的取值放松到[0,1]使$L(\boldsymbol{c} \odot \boldsymbol{w}; \mathcal{D})$对于$\boldsymbol{c}$可导：

$$\Delta L_j(w; \mathcal{D}) \approx g_j(w; \mathcal{D}) = \frac{\partial L(c \odot w; \mathcal{D})}{\partial c_j}\Big|_{c=1}$$

$g_j(w; \mathcal{D})$ 近似表示第j个参数对模型的贡献，归一化后作为最终的得分：

$$s_j = \frac{\left|g_j(\boldsymbol{w}; \mathcal{D})\right|}{\sum_{k=1}^{m} |g_k(\boldsymbol{w}; \mathcal{D})|}$$

根据稀疏度度控制要求$|\boldsymbol{c}|_0 \leq \kappa$，将得分$s_j$低的参数对应的mask$c_j$置为0，完

成模型剪枝，获得能耗较低的模型。由于该方法只需要计算Loss对mask $\boldsymbol{c}$的偏导数，而不要求精确的loss值，因此不需要对模型进行预训练，只需要极少的计算就可以完成剪枝任务。

以推荐模型中的嵌入矩阵为例，如果考虑特征选择问题，则需要将通用的剪枝问题构建改写如下：

$$min_{c,w} L(\boldsymbol{c} \odot \boldsymbol{w}; \mathcal{D}) = \min_{c,w} \frac{1}{n} \sum_{i=1}^{n} \mathrm{l}\big(\boldsymbol{c} \odot \boldsymbol{w}; (x_i, y_i)\big)$$

$$\text{s.t.}\ \boldsymbol{w} \in R^{m \times d}, \boldsymbol{c} \in \{0,1\}^m, |\boldsymbol{c}|_0 \leq \kappa$$

其中，$\boldsymbol{w}$是m个特征字段嵌入后（维度统一为d）拼接后的结果，实际上整个嵌入表征矩阵为$\boldsymbol{E} \in R^{M \times d}$，$M$ 是所有表征可能取值的个数之和。使用上述方法可以得到每个特征字段的得分s_j，去掉得分最小的若干特征字段且满足$|\boldsymbol{c}|_0 \leq \kappa$后（该场景下含义为最多选择$\kappa$个特征字段），我们得到被选中的$m'$个特征字段，并得到规模较小的嵌入表征矩阵$\boldsymbol{E}' \in R^{M' \times d}$。

方法有效性评估：

① 评估角度：本方案中提出了NAS搜索过程绿色化、搜索结果绿色化两个方向共四个方法，我们将从两个角度来评估方法的有效性：任务角度和系统角度。任务角度来说，我们将根据评估指标，单独评估每个方法的有效性；系统角度来说，我们将不限于该方案中提出的两个方向进行研究，最终将各个方向的方法结合，形成一整套低能耗、高性能的NAS系统，根据评估指标来测试最终系统的有效性。

② 评估指标：根据选题，我们需要从能耗、性能两个角度评估使用的方法，相对地使用两个角度的指标：能耗方面，对于搜索过程，我们将记录NAS算法产生最终结果所需要的时间，包括原始NAS算法以及优化后的NAS算法，以及搜索模型超图消耗的FLOPS以及参数量；对于搜索结果，我们将记录模型训练需要的时间、部署后产生预测结果需要的推断时间、最终模型消耗的FLOPS以及参数量；性能方面，我们将记录搜广营场景对应指标，如点击率预测场景中的AUC得分、LogLoss，排序场景中的NDCG、Recall Rate等。

③ 评估策略：我们将在具有不同数据分布特点的公开数据集以及搜广营真实业

务场景（如“有点东西”）中进行方法评估，得到客观、全面的结论。

技术优势：

① 创新性：现存的自动推荐系统仍然基于高能耗的NAS算法、剪枝方法设计，消耗大量的计算资源，具体来说，AutoLoss[16]、AutoField[9]、AutoDim[17]基于DARTS[15]设计，DeepLight[18]基于IMP（Iterative Magnitude Pruning，一种重复训练网络的剪枝方法[19]）设计。研发低能耗的自动推荐系统存在研究价值和探索空间；

② 可落地性：上述技术路线主要思想均较为直观，方便进行拓展、部署；

③ 有效性：使用上述技术路线可以损失极小模型效果的同时节约大量碳排放（模型搜索、部署），完成既定目标；

④ 高效性：上述技术路线本身带来的额外消耗远小于可以节约的NAS算法成本，效率高；

⑤ 易用性：上述技术路线方法设计完成后无需专家知识就可以拓展到多个业务，易于使用。

参考文献：

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[3]. Chen Bo, Zhao Xiangyu, Wang Yejing, Fan Wenqi, Guo Huifeng, and Ruiming Tang. 2022. Automated Machine Learning for Deep Recommender Systems: A Survey. (2022).

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[15]. Liu, Hanxiao, Karen Simonyan, and Yiming Yang. "Darts: Differentiable arch itecture search." arXiv:1806.09055 (2018).

[16]. Zhao, Xiangyu, et al. "Autoloss: Automated loss function search in recom mendations." Proceedings of the 27th ACM SIGKDD Conference on Knowle dge Discovery & Data Mining. 2021.

[17]. Zhao, Xiangyu, et al. "Autodim: Field-aware embedding dimension searchin recommender systems." Proceedings of the Web Conference 2021. 2021.

[18]. Deng, Wei, et al. "DeepLight: Deep lightweight feature interactions for ac celerating CTR predictions in ad serving." Proceedings of the 14th ACM int ernational conference on Web search and data mining. 2021.

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2. 研究计划

2.1. 项目期限：以实际合同签字生效日期为准，合同期建议 ***1*** 年。

2.2. 项目计划：

阶段	研究内容	时间	提交结果
1	合同签订，国内外现状调研	2022.12	调研报告书 1 份
2	系统初步设计	2023.1.1-2023.3.31	系统概要设计文档 1 份
3	系统实现与完善	2023.4.1-2023.6.30	详细设计文档 1 份，申请专利 1 个
4	系统完善	2023.7.1-2023.9.30	蚂蚁认可的 CCF-A 类高水平学术论文 2 篇
5	项目验收结题	2023.10.1-2023.12.31	原型系统软件 1 套、代码 1 份、其他验收材料；输送研究生到蚂蚁实习，完成项目落地

3. 预期成果

（1）算法原型：完成基于NAS的“绿色”搜推营模型自动搜索系统设计，并交付基于NAS的搜推营模型自动搜索系统以及源码；

（2）设计文档：提交项目概要设计文档和详细设计文档各1份；

（3）论文指标：发表蚂蚁认可的CCF-A类高水平学术论文2篇；

（4）专利指标：提交发明专利申请书底稿1份；

（5） 技术指标：与现有的搜推营模型相比，使用NAS搜索得到的模型能耗下降50%以上，模型效果下跌1%以内；

（6）业务指标：预计合作成果可落地到蚂蚁的“有点东西”业务中。

（7）提交验收测试报告1份；

（8）联合培养研究生2~4名，并输送公司实习。

4. 人力、设备等投入及项目预算

4.1. 人力投入（包括全部参与项目的人员）

人员姓名	所属机构与职务	人员工资（元/月）	参与项目时间(折合全时人月数)	人员工资总计（元）	承担职责	来蚂蚁实习时间（学生）
赵翔宇	香港城市大学 助理教授 副研究员	50000元/月	1	50000	PI	
王叶晶	香港城市大学博士生	15000元/月	4	60000	项目主要成员	是
刘启东	香港城市大学博士生	15000元/月	4	60000	项目主要成员	是
刘朗明	香港城市大学博士生	15000元/月	4	60000	项目主要成员	是
王茂林	香港城市大学博士生	15000元/月	2	30000	项目主要成员	

4.2. 设备投入

设备名称	数量	价格（元）	总计（元）	备注
GPU 服务器	3	0	0	现有设备，无需再次投入

4.3 项目预算

类别	总计（元）	备注
① **人员支出**	260000元	与4.1对应
② **研发支出**	50000元	
出差/会议费用	30000元	国际差旅15000元/趟*2趟，用于宣传研究成果
设备费	0元	现有设备，无需再次投入
其它费用	20000元	论文版面、专利申请费用
③ **直接费用**	310000元	①+②
④ **间接费用**	90000元	学校学院管理费**16.5%**，专票税费**默认6%**
总计	**400000**元	③+④

5. 教授个人简历

赵翔宇，香港城市大学(CityU)数据科学学院长聘轨助理教授、副研究员、博导。在此之前，他在美国密歇根州立大学获得博士学位，师从汤继良教授；在中国科技大学获得硕士学位，师从陈恩红教授。他目前的研究方向为人工智能、数据挖掘、机器学习及其在推荐系统（计算广告、搜索引擎和信息检索），城市计算（智慧城市和时空数据）等领域的应用，以及自动机器学习、深度强化学习、可信机器学习、多模态学习等。他在顶级会议(如KDD、WWW、SIGIR、AAAI、IJCAI、ICDE、ICDM、CIKM、WSDM、RecSys、ICLR)和期刊(如TOIS、SIGKDD、SIGWeb、EPL、APS)上发表了50多篇论文。他的研究获得了ICDM'21 Best-ranked Papers、全球人工智能华人新星奖、CCF-腾讯犀牛鸟基金、Criteo学者研究奖、Bytedance研究合作奖、MSU学位论文奖、以及AAAI/ACM SIGAI联合博士论文奖提名等奖项。他担任了超过100次顶级会议和期刊的(高级)程序委员或评审，和知名AI社区MLNLP (机器学习算法与自然语言处理)的创始学术委员会成员。他多次担任顶级会议KDD、WWW、SIGIR、IJCAI中研讨会组织者和教学讲座演讲者，获得大量关注和广泛好评。他曾在多家中美互联网企业从事研究工作，他的算法模型在多家公司上线，拥有和提交多项专利，并保持紧密合作关系。更多信息请访问https://zhaoxyai.github.io/。

注：项目负责人在基于AutoML的搜索推荐广告系统的研究和工程方面均有丰富经验：

1. 他在顶级会议KDD'22 [1]，ICLR'22 [2]，WWW'22 [3]，ICDM'21 [4]，KDD'21 [5]，WWW'21 [6]，SIGIR'20 [7] 发表七个篇此方向论文，其中6篇为一作或通讯作者；

2. 他在字节跳动合作项目(1年)、实习(1年)期间是基于AutoML的搜索推荐广告系统项目"SparseNAS"的主要成员，深度参与了SparseNAS项目的开发、上线过程。其中他领导了Embedding维度搜索模型[4]在字节跳动16个业务线上线(如头条、抖音、TikTok等)，节约了50%~70% Embedding参数量，显著提升了模型的训练和推断效率，并参与撰写了公司内部总结报告[8]；

3. 他在CCF-A类会议WWW'21、IJCAI'21、WWW'22做了基于AutoML的

搜索推荐广告系统的Tutorial学术讲座[9] ， 获得广泛关注和大量好评。并撰写AutoML推荐综述文章[10]。

[1] Adaptive Feature Selection in Deep Recommender Systems (KDD'2022) , CCF-A，通讯作者

[2] Automated Self-Supervised Learning for Graphs (ICLR'2022)

[3] Automating Feature Selection in Deep Recommender Systems (WWW'2022) , CCF-A，通讯作者

[4] AutoEmb: Automated Embedding Dimensionality Search in Streaming Recommendations (ICDM’2021) ，一作，Best-ranked Papers

[5] AutoLoss: Automated Loss Function Search in Recommendations (KDD'2021) , CCF-A，一作

[6] AutoDim: Field-aware Embedding Dimension Search in Recommender Systems (WWW'2021) , CCF-A，一作

[7] Automated Embedding Size Search in Deep Recommender Systems (SIGIR'2020) , CCF-A，共同一作

[8]坑都替你踩过了-- AutoML实验最佳实践

[9] https://advanced-recommender-systems.github.io/AutoML-Recommendations/

[10] Automated Machine Learning for Deep Recommender Systems: A Survey, 共同一作

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CONTACT INFORMATION

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POSITIONS

Assistant Professor (tenure-track), City University of Hong Kong — Sep 2021 - Present
School of Data Science

Associate Research Professor, City University of Hong Kong — Sep 2021 - Present
Shenzhen Research Institute

Director, City University of Hong Kong — Sep 2021 - Present
Applied Machine Learning (AML) Lab

Faculty Member, City University of Hong Kong — Oct 2021 - Present
Hong Kong Institute for Data Science

EDUCATION

Michigan State University

Doctor of Philosophy (Ph.D.) in Computer Science and Engineering — Jan 2017 - May 2021
- Advisor: Prof. Jiliang Tang
- Thesis: Adaptive and Automated Deep Recommender System
- Global Top 100 Chinese New Stars in Artificial Intelligence, Criteo Research Faculty Award, Bytedance Research Collaboration Award, MSU Dissertation Completion Fellowship, Nomination for Joint AAAI/ACM SIGAI Doctoral Dissertation Award (one per institution)

University of Science and Technology of China

Master of Science (M.S.) in Computer Science and Technology — Sep 2014 - Jun 2017
- Advisor: Prof. Enhong Chen
- Thesis: Exploring the Choice under Conflict for Social Event Participation
- Outstanding Master's Thesis Award (Top 15 in Anhui Province, China), Outstanding Graduate Award of USTC, National Scholarship for Graduate Students of China (Highest National Scholarship)

University of Electronic Science and Technology of China

Bachelor of Engineering (B.Eng.) in Software Engineering — Sep 2010 - Jun 2014
- Advisor: Prof. Tao Zhou and Prof. Ming Tang
- Thesis: Identifying Effective Multiple Spreaders by Coloring Complex Networks
- Outstanding Graduation Thesis Award (1% in university, 1/205 in department), Outstanding Graduate Award of UESTC

RESEARCH EXPERIENCE

Data Science and Engineering Lab, Michigan State University
Ph.D. Research Assistant — Jan 2017 - May 2021
- Advisor: Prof. Jiliang Tang
- Research Interests: Machine Learning, Information Retrieval, Urban Computing
- 20+ papers in top conferences (e.g., KDD, WWW, SIGIR, AAAI, ICDE, ICDM, CIKM, WSDM, RecSys) and journals (e.g., SIGKDD, SIGWeb)

Applied Machine Learning Research, Bytedance (US)
Research Intern — Jan 2021 - Apr 2021
- Mentor: Dr. Chong Wang
- Worked on AutoML for Recommendation/Advertising
- The outcomes were published in KDD'21, ICDM'21

Artificial Intelligence Foundation, Linkedin
Research Engineering Intern May 2020 - Aug 2020
- Mentors: Dr. Huiji Gao and Dr. Bo Long
- Worked on AutoML for Search/Recommendation/Advertising
- The outcomes were published in WWW'21

Applied Machine Learning, Bytedance
Research Intern Jun 2019 - Dec 2019
- Mentors: Dr. Chong Wang and Xiaobing Liu
- Worked on Reinforcement Learning and AutoML for Advertising/Recommendation
- The outcomes were published in KDD'20, SIGIR'20, AAAI'21

Advertising Department, Bytedance
Invited Visiting Scholar Sep 2018
- Mentor: Dr. Xiwang Yang
- Investigated Reinforcement Learning Based Computational Advertising

Data Science Lab, JD.com
Research Intern May 2018 - Aug 2018
- Mentor: Dr. Dawei Yin
- Worked on Reinforcement Learning Based Recommender System
- The outcomes were published in CIKM'20, SIGIR'20, WWW'21

Data Science Lab, JD.com
Research Intern Jun 2017 - Aug 2017
- Mentor: Dr. Dawei Yin
- Worked on Deep Learning Based Recommender System
- The outcomes were published in KDD'18, RecSys'18, ACM SIGWEB'19

Laboratory of Big Data Analysis and Application, University of Science and Technology of China
M.S. Research Assistant Sep 2014 - Dec 2016
- Advisor: Prof. Enhong Chen
- Research Interests: Data Mining, Urban Computing
- The outcomes were published in KDD'16, ICDM'17, DASFAA'17, KSEM'17

Big Data Research Center, University of Electronic Science and Technology of China
Undergraduate Research Assistant Sep 2011 - Jun 2014
- Advisors: Prof. Tao Zhou and Prof. Ming Tang
- Research Interests: Social Network and Complex Network Analysis, Complex System
- The outcomes were published in Europhysics Letters, Acta Physica Sinica

PUBLICATIONS

Tutorial and Workshop Proposals

- **Xiangyu Zhao**, Xin Xin, Weinan Zhang, Li Zhao, Dawei Yin, Grace Hui Yang, *"3rd Deep Reinforcement Learning for Information Retrieval: Fundamentals and Advances"*, 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2022**).
- **Xiangyu Zhao**, Wenqi Fan, Bo Chen, Ruiming Tang, *"Automated Machine Learning for Recommendations: Fundamentals and Advances"*, The Web Conference 2022 (**WWW'2022**).
- **Xiangyu Zhao**, Wenqi Fan, Dawei Yin, Jiliang Tang, *"Deep Recommender System: Fundamentals and Advances"*, The Web Conference 2021 (**WWW'2021**).
- Jiliang Tang, **Xiangyu Zhao**, Dawei Yin, Long Xia, Huiji Gao, Rui Chen, Jason Gauci, *"Deep Reinforcement Learning for Knowledge Discovery"*, The Web Conference 2021 (**WWW'2021**).
- Weinan Zhang, **Xiangyu Zhao**, Li Zhao, Dawei Yin, Grace Hui Yang, *"2nd Deep Reinforcement Learning for Information Retrieval: Fundamentals and Advances"*, 44th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2021**).

- Wenqi Fan, **Xiangyu Zhao**, Dawei Yin, Jiliang Tang, *"Deep Learning for Recommendations: Fundamentals and Advances"*, 30th International Joint Conference on Artificial Intelligence (**IJCAI'2021**).
- Weinan Zhang, **Xiangyu Zhao**, Li Zhao, Dawei Yin, Grace Hui Yang, Alex Beutel, *"Deep Reinforcement Learning for Information Retrieval: Fundamentals and Advances"*, 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2020**).
- Jiliang Tang, Dawei Yin, Long Xia, Alex Beutel, Minmin Chen, Shauki Jain, **Xiangyu Zhao**, *"Deep Reinforcement Learning for Knowledge Discovery"*, 25th ACM SIGKDD Conference on Knowledge Discovery and Data (**KDD'2019**).

Conference and Journal Papers

- Fengyi Song, Bo Chen, **Xiangyu Zhao***, Huifeng Guo, Ruiming Tang, *"AutoAssign: Automatic Shared Embedding Assignment in Streaming Recommendation"*, 22nd International Conference on Data Mining (**ICDM'2022**).
- Chenxu Zhu, Bo Chen, Huifeng Guo, Hang Xu, Xiangyang Li, **Xiangyu Zhao**, Weinan Zhang, Yong Yu, Ruiming Tang, *"AutoGen: An Automated Dynamic Model Generation Framework for Recommender System"*, 16th ACM International Conference on Web Search and Data Mining (**WSDM'2023**).
- Kesen Zhao, **Xiangyu Zhao***, Zijian Zhang, Muyang Li, *"Storage-saving Transformer for Sequential Recommendations"*, 31st ACM International Conference on Information and Knowledge Management (**CIKM'2022**).
- Pengfei He, Haochen Liu, **Xiangyu Zhao***, Hui Liu, Jiliang Tang, *"Personalized Probabilistic Strategic Parameter Optimization in Recommendations"*, 31st ACM International Conference on Information and Knowledge Management (**CIKM'2022**).
- Chi Zhang, Yantong Du, **Xiangyu Zhao***, Qilong Han, Rui Chen, Li Li, *"Hierarchical Item Inconsistency Signal learning for Sequence Denoising in Sequential Recommendation"*, 31st ACM International Conference on Information and Knowledge Management (**CIKM'2022**).
- Xinhang Li, Zhaopeng Qiu, **Xiangyu Zhao**, Zihao Wang, Yong Zhang, Chunxiao Xing, Xian Wu, *"Gromov-Wasserstein Guided Representation Learning for Cross-Domain Recommendation"*, 31st ACM International Conference on Information and Knowledge Management (**CIKM'2022**).
- Weilin Lin, **Xiangyu Zhao***, Yejing Wang, Tong Xu, Xian Wu, *"Adaptive Feature Selection in Deep Recommender Systems"*, 28th ACM SIGKDD Conference on Knowledge Discovery and Data (**KDD'2022**).
- Jingfan Chen, Wenqi Fan, Guanghui Zhu, **Xiangyu Zhao**, Chunfeng Yuan, Qing Li, Yihua Huang, *"Knowledge-enhanced Black-box Attacks for Recommendations"*, 28th ACM SIGKDD Conference on Knowledge Discovery and Data (**KDD'2022**).
- Zhi Zheng, Zhaopeng Qiu, Tong Xu, Xian Wu, **Xiangyu Zhao**, Enhong Chen, Hui Xiong, *"DDR: Dialogue based Doctor Recommendation for Online Medical Service"*, 28th ACM SIGKDD Conference on Knowledge Discovery and Data (**KDD'2022**).
- Muyang Li, **Xiangyu Zhao***, Chuan Lyu, Minghao Zhao, Runze Wu, Ruocheng Guo, *"MLP4Rec: A Pure MLP Architecture for Sequential Recommendations"*, 31st International Joint Conference on Artificial Intelligence (**IJCAI'2022**).
- Wenqi Fan, Xiaorui Liu, Wei Jin, **Xiangyu Zhao***, Jiliang Tang, Qing Li, *"Graph Trend Filtering Networks for Recommendation"*, 45th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2022**).
- Wei Jin, Xiaorui Liu, **Xiangyu Zhao**, Yao Ma, Neil Shah, Jiliang Tang, *"Automated Self-Supervised Learning for Graphs"*, 10th International Conference on Learning Representations (**ICLR'2022**).

- Haochen Liu, Da Tang, Ji Yang, **Xiangyu Zhao***, Hui Liu, Jiliang Tang, Youlong Cheng, *"Rating Distribution Calibration for Selection Bias Mitigation in Recommendations"*, The Web Conference 2022 (**WWW'2022**).
- Yejing Wang, **Xiangyu Zhao***, Tong Xu, Xian Wu, *"Automating Feature Selection in Deep Recommender Systems"*, The Web Conference 2022 (**WWW'2022**).
- Zhi Zheng, Zhaopeng Qiu, Tong Xu, Xian Wu, **Xiangyu Zhao**, Enhong Chen, Hui Xiong, *"CBR: Context Bias aware Recommendation for Debiasing User Modeling and Click Prediction"*, The Web Conference 2022 (**WWW'2022**).
- Zhi Zheng, Chao Wang, Tong Xu, Dazhong Shen, Penggang Qin, **Xiangyu Zhao**, Baoxing Huai, Xian Wu, Enhong Chen, *"Interaction-aware Drug Package Recommendation via Policy Gradient"*, ACM Transactions on Information and Systems (**TOIS**), 2022.
- **Xiangyu Zhao**, Wenqi Fan, Hui Liu, Jiliang Tang, *"Multi-type Urban Crime Prediction"*, 36th AAAI Conference on Artificial Intelligence (**AAAI'2022**).
- **Xiangyu Zhao**, Haochen Liu, Wenqi Fan, Hui Liu, Jiliang Tang, Chong Wang, Ming Chen, Xudong Zheng, Xiaobing Liu, Xiwang Yang, *"AutoEmb: Automated Embedding Dimensionality Search in Streaming Recommendations"*, 21st International Conference on Data Mining (**ICDM'2021**), **Best-ranked Papers**.
- **Xiangyu Zhao**, Haochen Liu, Wenqi Fan, Hui Liu, Jiliang Tang, Chong Wang, *"AutoLoss: Automated Loss Function Search in Recommendations"*, 27th ACM SIGKDD Conference on Knowledge Discovery and Data (**KDD'2021**).
- **Xiangyu Zhao**, *"Adaptive and Automated Deep Recommender Systems"*, **PhD Thesis**, extended abstract version at ACM SIGWEB Newsletter (**SIGWEB**).
- **Xiangyu Zhao**, Haochen Liu, Hui Liu, Jiliang Tang, Weiwei Guo, Jun Shi, Sida Wang, Huiji Gao, Bo Long, *"AutoDim: Field-aware Embedding Dimension Search in Recommender Systems"*, The Web Conference 2021 (**WWW'2021**).
- **Xiangyu Zhao**, Long Xia, Lixin Zou, Hui Liu, Dawei Yin, Jiliang Tang, *"UserSim: User Simulation via Supervised Generative Adversarial Network"*, The Web Conference 2021 (**WWW'2021**).
- **Xiangyu Zhao**, Changsheng Gu, Haoshenglun Zhang, Xiwang Yang, Xiaobing Liu, Jiliang Tang, Hui Liu, *"DEAR: Deep Reinforcement Learning for Online Advertising Impression in Recommender Systems"*, 35th AAAI Conference on Artificial Intelligence (**AAAI'2021**).
- Yingqiang Ge, Shuchang Liu, Ruoyuan Gao, Yikun Xian, Yunqi Li, **Xiangyu Zhao**, Changhua Pei, Fei Sun, Junfeng Ge, Wenwu Ou, Yongfeng Zhang, *"Towards Long-term Fairness in Recommendation"*, 13th International Conference on Web Search and Data Mining (**WSDM'2021**).
- Wenqi Fan, Tyler Derr, **Xiangyu Zhao**, Yao Ma, Hui Liu, Jianping Wang, Jiliang Tang, Qing Li, *"Attacking Black-box Recommendations via Copying Cross-domain User Profiles"*, 37th IEEE International Conference on Data Engineering (**ICDE'2021**).
- **Xiangyu Zhao**, Long Xia, Lixin Zou, Dawei Yin, Hui Liu, Jiliang Tang, *"Whole-Chain Recommendations"*, 29th ACM International Conference on Information and Knowledge Management (**CIKM'2020**).
- **Xiangyu Zhao**, Xudong Zheng, Xiwang Yang, Xiaobing Liu, Jiliang Tang, *"Jointly Learning to Recommend and Advertise"*, 26th ACM SIGKDD Conference on Knowledge Discovery and Data (**KDD'2020**).
- **Xiangyu Zhao***, Haochen Liu*, Chong Wang, Xiaobing Liu, Jiliang Tang, *"Automated Embedding Size Search in Deep Recommender Systems"*, 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2020**).

- Lixin Zou, Long Xia, Yulong Gu, **Xiangyu Zhao**, Weidong Liu, Jimmy Xiangji Huang, Dawei Yin, *"Neural Interactive Collaborative Filtering"*, 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2020**).
- **Xiangyu Zhao**, Liang Zhang, Long Xia, Zhuoye Ding, Dawei Yin, Jiliang Tang, *"Deep Reinforcement Learning for List-wise Recommendations"*, 1st Workshop on Deep Reinforcement Learning for Knowledge Discovery (**DRL4KDD'2019**).
- **Xiangyu Zhao**, Long Xia, Jiliang Tang, Dawei Yin, *"Deep Reinforcement Learning for Search, Recommendation, and Online Advertising: A Survey"*, ACM SIGWEB Newsletter (**SIGWEB**), 2019, Issue Spring: Article No.4.
- **Xiangyu Zhao**, Long Xia, Liang Zhang, Zhuoye Ding, Dawei Yin, Jiliang Tang, *"Deep Reinforcement Learning for Page-wise Recommendations"*, 12th ACM Recommender Systems Conference (**RecSys'2018**).
- **Xiangyu Zhao**, Liang Zhang, Zhuoye Ding, Long Xia, Jiliang Tang, Dawei Yin, *"Recommendations with Negative Feedback via Pairwise Deep Reinforcement Learning"*, 24th ACM SIGKDD Conference on Knowledge Discovery and Data (**KDD'2018**).
- **Xiangyu Zhao**, Jiliang Tang, *"Crime in Urban Areas: A Data Mining Perspective"*, ACM SIGKDD Explorations Newsletter (**SIGKDD Explorations**), 2018, 20(1): 1-12.
- **Xiangyu Zhao**, Jiliang Tang, *"Modeling Temporal-Spatial Correlations for Crime Prediction"*, 26th ACM International on Conference on Information and Knowledge Management (**CIKM'2017**).
- **Xiangyu Zhao**, Tong Xu, Yanjie Fu, Enhong Chen, Hao Guo, *"Incorporating Spatio-Temporal Smoothness for Air Quality Inference"*, 17th International Conference on Data Mining (**ICDM'2017**).
- **Xiangyu Zhao**, Jiliang Tang, *"Exploring Transfer Learning for Crime Prediction"*, 17th International Conference on Data Mining (**ICDM'2017**), Ph.D. Forum.
- Tong Xu, Hengshu Zhu, **Xiangyu Zhao**, Qi Liu, Hao Zhong, Enhong Chen, Hui Xiong, *"Taxi Driving Behavior Analysis in Latent Vehicle-to-Vehicle Networks: A Social Influence Perspective"*, 22nd ACM SIGKDD Conference on Knowledge Discovery and Data (**KDD'2016**).
- **Xiangyu Zhao**, Tong Xu, Qi Liu, Hao Guo, *"Exploring the Choice under Conflict for Social Event Participation"*, 21st International Conference on Database Systems for Advanced Applications (**DASFAA'2016**).
- Hao Guo, Xin Li, Ming He, **Xiangyu Zhao**, Guiquan Liu, Guandong Xu, *"CoSoLoRec: Joint Factor Model with Content, Social, Location for Heterogeneous Point-of-Interest Recommendation"*, 9th International Conference on Knowledge Science, Engineering and Management (**KSEM'2016**).
- **Xiangyu Zhao**, Bin Huang, Ming Tang, Haifeng Zhang, Duanbing Chen, *"Identifying Effective Multiple Spreaders by Coloring Complex Networks"*, Europhysics Letters (**EPL**), 2015, 108(6): 68005. (SCI IF: 2.095)
- Bin Huang, **Xiangyu Zhao**, Kai Qi, Ming Tang, Younghae Do, *"Coloring the Complex Networks and its Application for Immunization Strategy"*, Acta Physica Sinica (**APS**), 2013, 62(21): 218902-218902. (SCI IF: 0.845)

PROJECTS

Unified Multi-modal Multi-task Learning Paradigm for Smart Cities
- **PI**: HKIDS Early Career Research Grant (9360163), 11/22-10/24, 500,000 HKD

Wise-waste System Integration Development and Policies Applications in Hong Kong
- **Co-PI**: SIRG-CityU Strategic Interdisciplinary Research Grant (7020046), 05/22-10/24, 300,000 HKD

A Social Media Powered Preseason Production Planning System for Fast-fashion New Releases
- **Co-PI**: SIRG-CityU Strategic Interdisciplinary Research Grant (7020074), 05/22-10/24, 300,000 HKD

Developing Largescale Streaming Recommender System via Deep Reinforcement Learning Techniques
- **PI**: APRC - CityU New Research Initiatives (CityU Start-up Fund 9610565), 03/22-02/24, 1,113,600 HKD

Personalized Medical Content Recommendation System based on Multimodal and AutoML
- **PI**: CCF-Tencent Open Fund (9229093), 11/21-12/23, 261,725 HKD

Adaptive and Automated Deep Recommender Systems
- **PI**: MSU Dissertation Completion Fellowship, 2020

Deep Reinforcement Learning for Computational Advertising
- **Co-I**: Bytedance Research Collaboration Award, 2018

Deep Reinforcement Learning for Capturing Real-Time Feedback
- **Co-I**: Criteo Faculty Research Award, 2018

Identifying Effective Multiple Spreaders by Coloring Complex Networks
- **PI**: Ginkgo Innovation and Entrepreneurship Fund, 2013, outstanding completed project

Epidemic spreading on multi-relational networks
- **Co-I**: Ginkgo Innovation and Entrepreneurship Fund, 2012, outstanding completed project

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TEACHING EXPERIENCE

DLP Workshop @ KDD'22
Keynote Talk Aug 2022
- Topic: *"Automated Machine Learning for Recommendations: Fundamentals and Advances"*

East China Normal University
Invited Talk May 2022
- Topic: *"Deep Reinforcement Learning based Recommendations"*

Shein
Invited Talk May 2022
- Topic: *"Adaptive and Automated Deep Recommender Systems"*

Harbin Engineering University
Invited Talk May 2022
- Topic: *"Adaptive and Automated Deep Recommender Systems"*

The Web Conference 2022 (WWW'22)
Conference Tutor Apr 2022
- Tutorial: *"Automated Machine Learning for Recommendations: Fundamentals and Advances"*

City University of Hong Kong
Lecturer Jan 2022 - Apr 2022
- Course: *"SDSC8007 - Deep Learning"*, for graduate students

Bytedance
Invited Talk Dec 2021
- Topic: *"Adaptive and Automated Deep Recommender Systems"*

National University of Singapore
Invited Talk Dec 2021
- Topic: *"Deep Reinforcement Learning based Recommendations"*

International Joint Conference on Artificial Intelligence 2021 (IJCAI'21)
Conference Tutor Aug 2021
- Tutorial: *"Deep Learning for Recommendations: Fundamentals and Advances"*

The Web Conference 2021 (WWW'21)
Conference Tutor Apr 2021
- Tutorial: *"Deep Recommender System: Fundamentals and Advances"*

Noah's Ark Lab, Huawei
Invited Talk Dec 2020
- Topic: *"Automated Machine Learning (AutoML) based Recommendations"*

East China Normal University
Guest Lecturer Nov 2019
- Topic: *"Urban Security and Crime Prediction"*

Shandong University
Guest Lecturer Sep 2019
- Topic: *"Deep Reinforcement Learning for Online Advertising"*

TAL Education Group
Invited Talk Aug 2019
- Topic: *"Deep Reinforcement Learning based Recommendations"*

Beijing University of Chemical Technology
Guest Lecturer Jul 2019
- Topic: *"Urban Security and Crime Prediction"*

Leiphone Online Course
Lecturer Nov 2018
- Course: *"Deep Reinforcement Learning for E-commerce"*

Kuaishou.com
Invited Talk Aug 2018
- Topic: *"Deep Reinforcement Learning based Recommendations"*

University of Science and Technology of China
Teaching Assistant Sep 2015 - Jan 2016
- Course: *"Machine Learning and Knowledge Discovery"*, for graduate students

University of Electronic Science and Technology of China
Teaching Assistant Jan 2013 - Jun 2013
- Course: *"Computer Science Research"*, for undergraduate students

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SERVICES

Organizer
- 3rd Workshop on Reinforcement Learning for Information Retrieval@SIGIR'22 Jul 2022
  One of the most popular workshops at SIGIR'22 with more than 300 attendees
- MLNLP: Founding academic committee members Oct 2021
  The largest AI community in China with 800,000 followers
- 2nd Workshop on Reinforcement Learning for Information Retrieval@SIGIR'21 Jul 2021
  One of the most popular workshops at SIGIR'21 with more than 200 attendees
- 2nd Workshop on Deep Reinforcement Learning for Knowledge Discovery@WWW'21 Apr 2021
  One of the most popular workshops at WWW'21 with more than 200 attendees
- 27th International Conference on Neural Information Processing (ICONIP'20) Nov 2020
  Session Chair of Natural Language Processing Session
- Workshop on Reinforcement Learning for Information Retrieval@SIGIR'20 Jul 2020
  One of the most popular workshops at SIGIR'20 with more than 200 attendees
- Workshop on Deep Reinforcement Learning for Knowledge Discovery@KDD'19 Aug 2019
  One of the most popular workshops at KDD'19 with more than 500 attendees

Conference Senior Program Committee Member

- AAAI Conference on Artificial Intelligence (AAAI) 2023
- The Web Conference (WWW) 2023
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2022
- ACM International Conference on Web Search and Data Mining (WSDM) 2022
- International Joint Conferences on Artificial Intelligence (IJCAI) 2021

Conference Program Committee Member

- International Conference on Learning Representations (ICLR) 2020-2023
- AAAI Conference on Artificial Intelligence (AAAI) 2020-2022
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2021-2022
- International Joint Conferences on Artificial Intelligence (IJCAI) 2020-2022
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR) 2022
- SIAM International Conference on Data Mining (SDM) 2022
- International Conference on Machine Learning (ICML) 2021-2022
- Conference on Neural Information Processing Systems (NeurIPS) 2021-2022
- Conference on Information and Knowledge Management (CIKM) 2020-2022
- ACM Conference on Recommender Systems (RecSys) 2019-2022
- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD) 2022
- IEEE International Conference on Big Data (BigData) 2018-2022
- European Conference on Information Retrieval (ECIR) 2021
- Conference on Neural Information Processing (ICONIP) 2020
- Reinforcement Learning for Real Life@ICML'19 2019-2021
- Workshop on Artificial Intelligence for Education@AAAI'20 2020
- Workshop on Industrial Recommendation Systems@KDD'20 2020
- Conference on Natural Language Processing and Chinese Computing (NLPCC) 2019

Conference Sub-Reviewer

- The Web Conference (WWW) 2018-2021
- ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR) 2017-2021
- Annual Meeting of the Association for Computational Linguistics (ACL) 2021
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2016-2020
- ACM International Conference on Web Search and Data Mining (WSDM) 2018-2020
- International Conference on Web and Social Media (ICWSM) 2017-2020
- AAAI Conference on Artificial Intelligence (AAAI) 2017-2019
- Conference on Information and Knowledge Management (CIKM) 2016-2019
- Conference on Empirical Methods in Natural Language Processing (EMNLP) 2019
- Conference on Advances in Social Networks Analysis and Mining (ASONAM) 2015-2019
- ACM Conference on Recommender Systems (RecSys) 2016-2018
- ACM CHI Conference on Human Factors in Computing Systems (CHI) 2018
- SIAM International Conference on Data Mining (SDM) 2017
- Conference on Database Systems for Advanced Applications (DASFAA) 2016-2017
- IEEE International Conference on Data Mining (ICDM) 2016
- Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD) 2016
- APWeb-WAIM International Joint Conference on Web and Big Data (WAIM) 2016
- Conference on Knowledge Science, Engineering and Management (KSEM) 2015-2016

Journal Reviewer

- IEEE Transactions on Knowledge and Data Engineering (IEEE TKDE) 2019-2021
- ACM Transactions on Knowledge Discovery from Data (ACM TKDD) 2018-2021
- ACM Transactions on Information Systems (ACM TOIS) 2020-2021
- ACM Computing Surveys (ACM CSUR) 2021
- Machine Learning 2020
- Journal of Computer Science and Technology (JCST) 2020
- ISPRS International Journal of Geo-Information (IJGI) 2020
- Electronics and Telecommunications Research Institute (ETRI) 2019-2020
- IEEE Access 2019-2020
- IEEE Intelligent Systems 2019
- Journal of Big Data 2019
- Sustainability 2021
- Information Processing and Management 2021

Volunteer

- AAAI Conference on Artificial Intelligence (AAAI) 2021
- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2018-2020
- ACM Recommender Systems (RecSys) 2018
- Chinese Conference on Complex Networks (CCCN) 2011

ACADEMIC HONORS & AWARDS

Assistant Professor (tenure-track) at CityU

- The Gold Medal and Top 10 best innovation award, the 7th International Invention Innovation Competition in Canada (iCAN 2022), supervisor 2022
- The Gold Award with Grand Prize (US$13,000) and Microsoft Outstanding AI Influencer Award, the Global AI Challenge for Building E&M Facilities, supervisor 2022
- The 3rd place in Carbon Neutral Competition track, the 3rd Shandong Province Data Application Innovation and Entrepreneurship Competition, supervisor 2022
- Start-up Fund of CityU (PI) 2022
- ICDM 2021 Best-ranked Papers Award 2021
- CCF-Tencent Open Fund (PI) 2021

Doctor of Philosophy (Ph.D.) at MSU

- Nomination for Joint AAAI/ACM SIGAI Doctoral Dissertation Award (one per institution) 2021
- Global Top 100 Chinese New Stars in Artificial Intelligence 2021
- WWW 2021 Student Scholarship Award 2021
- AAAI 2021 Student Travel Award 2021
- MSU Dissertation Completion Fellowship (PI) 2021
- CIKM 2020 Student Travel Award 2020
- KDD 2020 Student Travel Award 2020
- KDD 2019 Student Travel Award 2019
- Bytedance Research Award (Co-I, based on my research) 2018
- RecSys 2018 Student Travel Award 2018
- KDD 2018 Student Travel Award 2018
- Criteo Research Award (Co-I, based on my research) 2018
- SDM 2018 Student Travel Award 2018
- CIKM 2017 Student Travel Award 2017

Master of Science (M.S.) at USTC

- Outstanding Master's Thesis Award (Top 15 in Anhui Province) 2018
- Outstanding Graduate Award of USTC 2017
- National Scholarship for Graduate Students of China (Highest National Scholarship) 2017
- Graduate Student First-class Academic Scholarship 2016
- Graduate Student First-class Academic Scholarship 2015
- Graduate Student First-class Academic Scholarship 2014

Bachelor of Engineering (B.Eng.) at UESTC

- Outstanding Graduation Thesis (1% in university, 1/205 in department) 2014
- Outstanding Graduate Award of UESTC 2014
- Ginkgo Innovation and Entrepreneurship Fund (PI, outstanding completed project) 2013
- First Prize of China Contemporary Undergraduate Mathematical Contest in Modeling (Sichuan) 2012
- First Prize of Mathematical Contest in Modeling of UESTC (3/200, only Freshman Team) 2012
- Ginkgo Innovation and Entrepreneurship Fund (Co-I, outstanding completed project) 2012
- Futong Scholarship (3/600) 2011

BIOGRAPHY

Dr. Xiangyu Zhao is an assistant professor of the school of data science at City University of Hong Kong (CityU). Prior to CityU, he completed his PhD (2021) at MSU, MS (2017) at USTC and BEng (2014) at UESTC. His current research interests include data mining and machine learning, especially (1) Personalization, Recommender System, Online Advertising, Search Engine, and Information Retrieval; (2) Urban Computing, Smart City, and GeoAI; (3) Deep Reinforcement Learning, AutoML, Trustworthy AI and Multimodal ML; and (4) AI for Social Computing, Finance, Education, Ecosystem, and Healthcare. He has published more than 50 papers in top conferences (e.g., KDD, WWW, AAAI, SIGIR, IJCAI, ICDE, CIKM, ICDM, WSDM, RecSys, ICLR) and journals (e.g., TOIS, SIGKDD, SIGWeb, EPL, APS). His research has been awarded ICDM'21 Best-ranked Papers, Global Top 100 Chinese New Stars in AI, CCF-Tencent Open Fund, Criteo Research Award, Bytedance Research Award, MSU Dissertation Fellowship, and nomination for Joint AAAI/ACM SIGAI Doctoral Dissertation Award. He serves as top data science conference (senior) program committee members and session chairs (e.g., KDD, AAAI, WWW, IJCAI, ICML, ICLR, CIKM), and journal guest editors and reviewers (e.g., TKDE, TKDD, TOIS, TIST, CSUR, Frontiers in Big Data). He serves as the organizers of DRL4KDD@KDD'19, DRL4IR@SIGIR'20, 2nd DRL4KD@WWW'21, 2nd DRL4IR@SIGIR'21, 3rd DRL4IR@SIGIR'22 and a lead tutor at WWW'21/22 and IJCAI'21. He also serves as the founding academic committee members of MLNLP, the largest AI community in China with 800,000 members/followers. The models and algorithms from his research have been launched in the online system of many companies. Please find more information at https://zhaoxyai.github.io/.